Rakamin Academy X kimia farma

Kimia Farma Performance Analytics

Business Performance 2020-2023

Presented by

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Mathematics Graduate | Aspiring Data Analyst

Mathematics graduate with a **strong analytical background** and **practical experience** in data analytics training and projects. Skilled in Python, SQL, Excel, Power BI, and Looker Studio, with a focus on analyzing data, identifying trends, and translating findings into **actionable business insights**.

Bachelor of Mathematics - 2026
Universitas Sebelas Maret (GPA 3.64/4.00)

Data Analytics and Generative AI Program - 2024
RevoU Tech Academy | MSIB Batch 7

Technical Tools |

Courses and Certification

Data Analytics and Gen AI **RevoU Tech Academy (MSIB Batch 7)** Final Score: 96/100	View Certificate	2024
Data Science Foundation **Great Learning**	View Certificate	2024
Belajar Dasar Visualisasi Data **Dicoding Indonesia**	View Certificate	2024
Data Science Fundamentals **DQLab**	View Certificate	2024
Bootcamp Data Analyst with Excel **DQLab**	View Certificate	2023

About Company

PT Kimia Farma Tbk is an Indonesian pharmaceutical and healthcare company engaged in pharmaceutical manufacturing, distribution, retail, and healthcare services. As part of the Bio Farma Group, the company operates an extensive healthcare network across Indonesia.

Vision

To become the preferred integrated healthcare company that creates sustainable value.

Mission

1. To develop businesses across the pharmaceutical industry, distribution, retail, and healthcare services.
2. To uphold good corporate governance and operational excellence through professional human resources.
3. To create added value and benefits for all stakeholders.

Project Portfolio

This project analyzes **Kimia Farma's business performance from 2020 to 2023** using four datasets covering transactions, inventory, branches, and products. The datasets were consolidated in **BigQuery** and visualized in **Looker Studio** to assess annual sales trends, estimated net profit, provincial contributions, and customer ratings.

Problem Statement.

How effectively did Kimia Farma perform across sales, estimated profitability, and customer experience from 2020 to 2023, and what **performance gaps** and **improvement opportunities** can be identified across its provinces and branches?

Project Documentation: Explore on GitHub

Importing Dataset to BigQuery

The four CSV files provided by Kimia Farma were uploaded to Google BigQuery. This process involved three main steps:

Creating the BigQuery Project

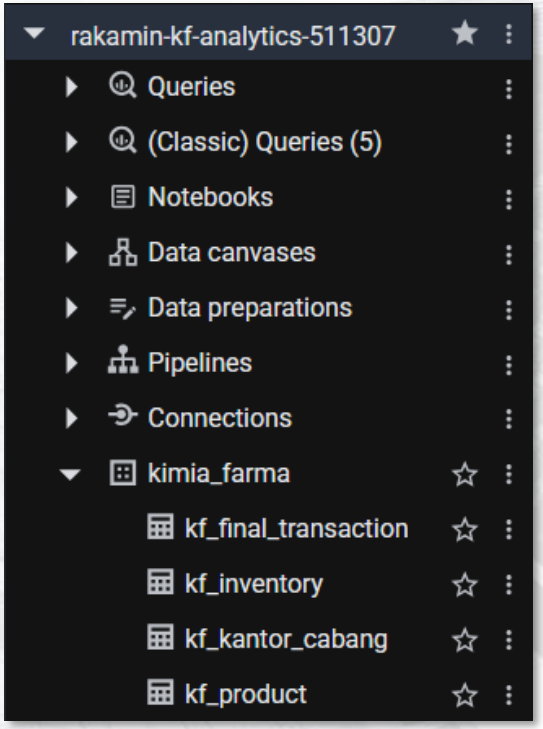
The project was named ***rakamin-kf-analytics-511307*** to serve as the workspace for the analysis.

Creating the Dataset

A dataset named ***kimia_farma*** was created to store the imported tables.

Creating Tables and Uploading CSV Files

Four tables were created by importing the provided CSV files. The tables are ***kf_final_transaction***, ***kf_inventory***, ***kf_kantor_cabang***, and ***kf_product***. Each table contains transaction, inventory, branch, or product data for further analysis.



Final Analysis Table

rakamin-kf-analytics-511307 / Datasets / kimia_farma / Tables / tabel_analisa

tabel_analisa | Query | Chat | Open in | Share | Copy | Snapshot | Delete | Export / sync | Refres

Schema | Details | Preview | Insights | Lineage | Data Profile | Data Quality

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id	product_name	actual_price	discount_per...	persentase_...	nett_sales	nett_profit	rating_transa...
1	TRX5758357	2023-07-20	83742	Kimia Farma - Apotek	Pematangsiantar	Sumatera Utara	4.8	Robin Brown	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.1
2	TRX6978726	2022-05-26	93878	Kimia Farma - Apotek	Semarang	Jawa Tengah	4.5	Suzanne Miller	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.5
3	TRX4949710	2023-04-02	65948	Kimia Farma - Apotek	Tarakan	Kalimantan Utara	4.4	Kathryn Mccarthy	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	4.9
4	TRX8801605	2020-10-16	96598	Kimia Farma - Apotek	Denpasar	Bali	4.7	Andres Campos	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	4.6
5	TRX5046417	2022-03-22	42820	Kimia Farma - Apotek	Mataram	Nusa Tenggara Barat	4.1	Andrew Bates	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.7
6	TRX5651355	2023-12-26	14493	Kimia Farma - Apotek	Gorontalo	Gorontalo	4.0	James Patterson	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.2
7	TRX3719390	2020-05-24	63815	Kimia Farma - Apotek	Sukabumi	Jawa Barat	4.7	Elizabeth Gutierrez	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	4.6
8	TRX7511418	2020-05-31	12802	Kimia Farma - Apotek	Garut	Jawa Barat	4.1	Brandi Martin	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.1
9	TRX3291498	2021-11-24	57026	Kimia Farma - Apotek	Pematangsiantar	Sumatera Utara	4.1	Jenny Miranda	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.5
10	TRX6938546	2021-06-25	47503	Kimia Farma - Apotek	Denpasar	Bali	4.1	Max Sutton	KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	178.5	3.9

The final analysis table, named ***tabel_analisa***, was created in BigQuery by integrating transaction records with branch and product information. It contains key transaction details, including customer names, branch ratings, transaction ratings, and financial metrics such as net sales and net profit.

BigQuery Syntax

```sql
CREATE OR REPLACE TABLE `rakamin-kf-analytics-511307.kimia_farma.tabel_analisa` AS

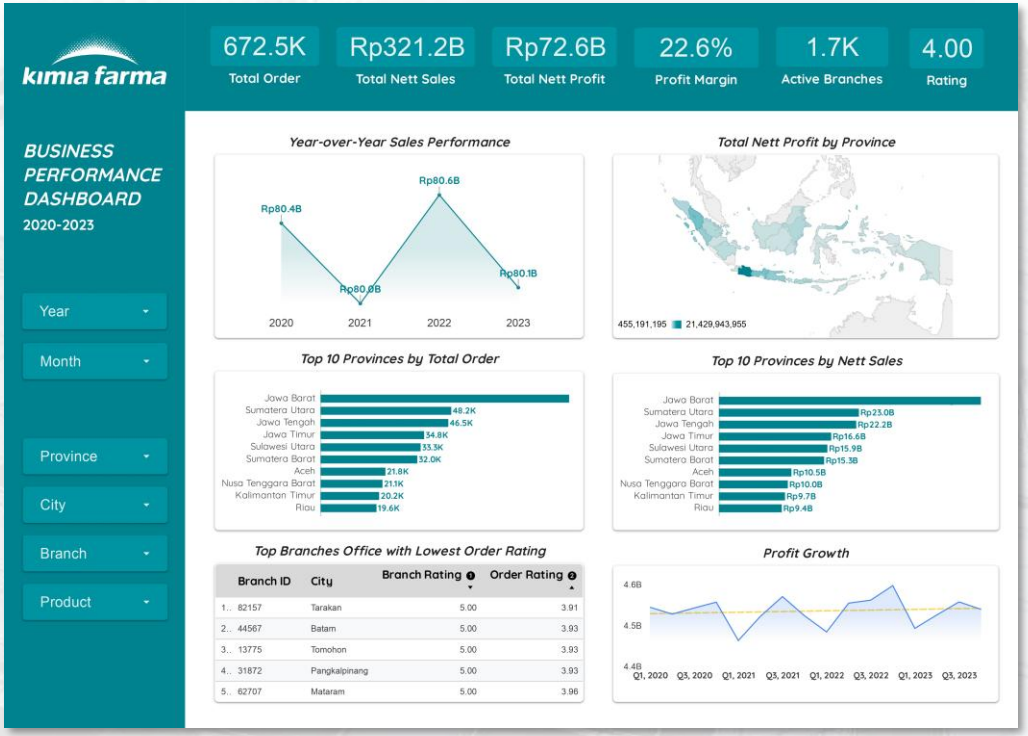
-- Step 1. Data integration
WITH joined AS (
  SELECT
    t.transaction_id,
    t.date,
    t.branch_id,
    b.branch_name,
    b.kota,
    b.provinsi,
    b.rating AS rating_cabang,
    t.customer_name,
    t.product_id,
    p.product_name,
    t.price AS actual_price,
    t.discount_percentage,
    t.rating AS rating_transaksi
  FROM `rakamin-kf-analytics-511307.kimia_farma.kf_final_transaction` AS t
  LEFT JOIN `rakamin-kf-analytics-511307.kimia_farma.kf_kantor_cabang` AS b
    ON t.branch_id = b.branch_id
  LEFT JOIN `rakamin-kf-analytics-511307.kimia_farma.kf_product` AS p
    ON t.product_id = p.product_id
),

-- Step 2. Metric calculation
calculated AS (
  SELECT
    *,
    CASE
      WHEN actual_price <= 50000  THEN 0.10
      WHEN actual_price <= 100000 THEN 0.15
      WHEN actual_price <= 300000 THEN 0.20
      WHEN actual_price <= 500000 THEN 0.25
      ELSE 0.30
    END AS persentase_gross_laba,
    actual_price * (1 - discount_percentage) AS nett_sales
  FROM joined
)

-- Step 3. Final analysis table
SELECT
  transaction_id,
  date,
  branch_id,
  branch_name,
  kota,
  provinsi,
  rating_cabang,
  customer_name,
  product_id,
  product_name,
  actual_price,
  discount_percentage,
  persentase_gross_laba,
  nett_sales,
  (actual_price * persentase_gross_laba)
    - (actual_price - nett_sales) AS nett_profit,
  rating_transaksi
FROM calculated;
```

The SQL query uses **LEFT JOIN** operations to integrate transaction records with branch and product details. It applies gross profit margin rates based on actual product prices, calculates net sales after discounts, and derives estimated net profit by deducting the discount amount from the expected gross profit. The resulting analysis table provides the financial and operational metrics used for further business performance analysis.

Dashboard Performance Analytics



The dashboard summarizes Kimia Farma's business performance from **2020 to 2023**, covering sales, estimated profitability, provincial contributions, and customer ratings. Key metrics include **Rp321.17B in net sales**, **Rp72.56B in estimated net profit**, and an **estimated profit margin of 22.6%**. Interactive filters enable performance comparisons across reporting periods and regions.

Interactive Dashboard: [Explore on Looker Studio]

Insights

No.	Insights
1	**Stagnant Sales Growth.** Annual net sales remained stable at approximately Rp80B, with YoY growth ranging from -0.57% to +0.68%
2	**Quarterly Profit Fluctuations.** Q1 recorded the lowest quarterly estimated profit in 2021–2023, indicating recurring quarterly fluctuations
3	**Provincial Sales Concentration.** Jawa Barat contributed 29.54% of net sales, followed by Sumatera Utara and Jawa Tengah. The top three provinces accounted for 43.62%
4	**Customer Rating Discrepancies.** 96 branches with perfect ratings (5.00) recorded transaction ratings below 4.00

Business Recommendations

No.	Business Recommendation
1	**Strengthen Sales Performance.** Implement targeted promotions, product bundles, and loyalty programs to support sustainable sales growth
2	**Optimize Quarterly Profitability.** Review Q1 demand, discount patterns, and inventory allocation to improve quarterly profitability
3	**Expand Regional Market Opportunities.** Strengthen leading markets while pursuing targeted growth opportunities across other provinces
4	**Improve Customer Experience.** Investigate rating discrepancies and strengthen service standards, staff training, and complaint resolution

Thank You

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